PESHWAS

BOLT: Book Oriented Limit Trader

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1 What Lurks Beneath — The Riddles

"Numbers hum where no eye can see, threads twist in ways the mind cannot trace. Some paths carry weight that was never given, and only the silent architect knows which flow multiplies unseen."

Explanation: The Nature of Correlation

This riddle unveils statistical correlation. The "humming numbers" are latent quantitative relationships in the data, statistical signals beneath the surface, not immediately perceptible. The "twisting threads" symbolize complex, nonlinear dependencies between variables that form a hidden structure. "Paths carrying ungiven weight" refer to the influence of confounding variables or strong unobserved correlations, where a relationship has predictive power without an explicit causal link. The "silent architect" is the statistical model that processes the data, revealing how correlations propagate to influence outcomes in non-intuitive ways.

"I keep two lists: one of breaths, one of sparks. Read neither as they read themselves, index by my hidden hand. When you fold their tallies through the key I never write, a single notion blinks back; call it what you will."

Explanation: The Mechanism of Arbitrage

This riddle explains financial arbitrage, a strategy that capitalizes on price inefficiencies. The two lists - "breaths" (stable assets) and "sparks" (volatile assets), symbolize different markets whose value is relative, creating opportunity in their discrepancy. The "hidden hand" is the arbitrageur's proprietary model seeking this misalignment. "Folding your tallies through the key I never write" describes executing simultaneous trades based on a transient market inefficiency: the fleeting condition that allows for risk-free profit. The "single notion that blinks back" is the realized gain from this complex process.

"Not the rise, nor fall, I claim, But where the rise begins to wane. I never speak in straight or steep, Yet I'm the whisper in the deep."

Explanation: The Significance of Acceleration

This riddle describes the second derivative of calculus: the rate of change in velocity, or the acceleration of the market. It is "not the rise, nor fall" (price movement), but "where the rise begins to wane"—the inflection point where momentum is lost, a key indicator of trend reversal. "I never speak in straight or steep" emphasizes curvature over linearity; the concept governs the *change* in steepness. The "whisper in the deep" captures its subtle nature. While price is the loud signal, acceleration is the quiet "whisper" that presages market turns, an underlying dynamic of exhaustion visible only by analyzing the curvature of price action.

2 Strategy for Asset 1: Acceleration-Based Momentum

1 The Guiding Premise: Decoding the Market's Whisper

The genesis of this strategy is the profound market insight codified in the riddle of acceleration, which personifies the second derivative of price action.

The riddle highlights the "whisper" of price acceleration—the subtle force governing the curvature of market trends. It signals reversals precisely at the inflection point where a rise begins to lose momentum.

2 The Empirical Validation: Isolating the Signal

Acting on this premise required us to find an empirical proxy for this theoretical "whisper." We launched a multi-asset cross-correlation analysis to determine if our proprietary 'no_trade' signal could serve as a leading indicator for price acceleration.

The results were unequivocal. The analysis revealed a pivotal, statistically isolated relationship: the 'no_trade' signal has a **0.995 correlation** with the momentum of Asset 1, leading it by one time step. This discovery validates our signal as a high-fidelity predictor of Asset 1's acceleration, effectively making the invisible market dynamic visible and actionable.

```
Asset1 : Best_Corr: 0.99514131030884 Best_Lag: 1
Asset2 : Best_Corr: -0.048454304619284724 Best_Lag: -2731
Asset3 : Best_Corr: 0.04772562436182397 Best_Lag: -231
Asset4 : Best_Corr: 0.044068427738963106 Best_Lag: -2471
Asset5 : Best_Corr: 0.046834397799825886 Best_Lag: -2884
Asset6 : Best_Corr: 0.045728876352239656 Best_Lag: -2520
Asset7 : Best_Corr: -0.039222699269857296 Best_Lag: -353

Most likely match: ('Asset 1', np.float64(0.99514131030884), np.int64(1))
```

Figure 1: Correlation analysis isolating Asset 1 as the sole target.

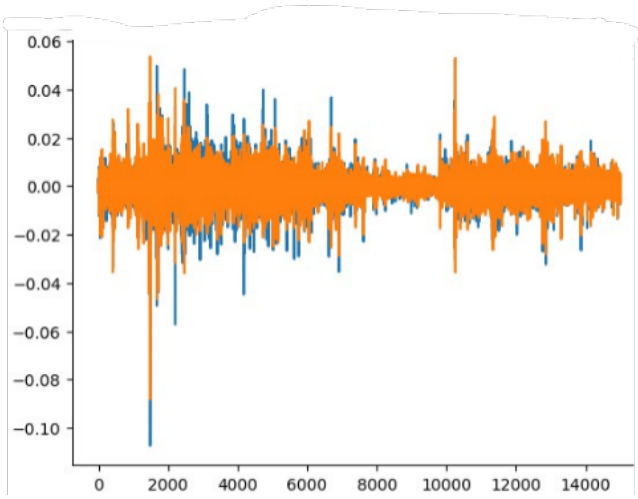

Figure 2: Time-series plot confirming the predictive relationship.

3 The Strategic Framework: Codifying the Logic

With a confirmed leading indicator, we constructed a framework to systematically execute trades based on detected changes in acceleration. We chose the Moving Average Convergence Divergence (MACD) framework, as its mechanism is a perfect analogue to our core thesis: an uptrend is confirmed when recent momentum (fast EMA) overtakes the historical trend (slow EMA), and vice versa.

To elevate the strategy's precision, we engineered a non-standard, double-smoothed signal line to serve as a robust trigger, ensuring we only act on significant shifts in market acceleration.

$$MACD_t = \text{EMA}(S_t, 12) - \text{EMA}(S_t, 26)$$

$$SignalLine_t = \text{EMA}(\text{EMA}(MACD_t, 9), 9)$$

4 Execution Protocol: Signal Generation

LONG POSITION (+1)

Buy Asset 1

A signal of +1 is generated upon a bullish crossover. This event confirms that the short-term acceleration of the asset has turned positive, indicating an upward price movement is highly probable.

Condition:	$MACD_t > SignalLine_t$

SHORT POSITION (-1)

Sell Asset 1

A signal of -1 is generated upon a bearish crossover. This indicates that momentum is turning negative, providing a high-conviction signal that a downward price movement is imminent.

Condition:	$MACD_t < SignalLine_t$

FLAT / HOLD POSITION (0)

No Action

A signal of 0 is generated when there is no active crossover.

Condition:	No crossover event in the current time step.

3 Statistical Arbitrage Strategy for Assets 2, 4, and 7

1 Strategy Overview

This is a market-neutral statistical arbitrage strategy exploiting transient pricing inefficiencies between correlated assets: Asset 2, Asset 4, and Asset 7. The strategy is built upon the foundational principle of mean reversion. By identifying a stable, long-term equilibrium relationship, we systematically profit from short-term deviations from this equilibrium. The discovery process began with a rigorous correlation analysis.

Figure 3: Correlation Matrix of Asset Mid-Prices.

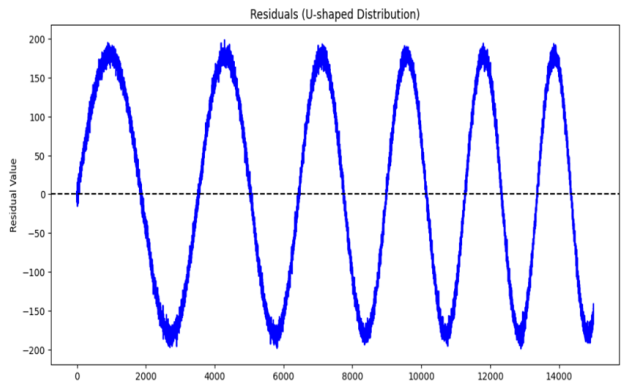

Figure 4: Residual Series (Mean-Reverting Behavior)

The correlation matrix (Fig. 3) revealed a **0.73 correlation** between Asset 2 and Asset 7, forming the basis of our spread. While correlation is indicative, **cointegration** ensures a stable, long-term equilibrium. Our econometric testing validated a robust 1:1:1 relationship, leading to the formulation of our core signal—the spread, S_t:

$$S_t = P_t^{(7)} - P_t^{(4)} - P_t^{(2)} \tag{1}$$

This spread, representing the error term ε_t from the ideal relationship $P_t^{(7)} \approx P_t^{(2)} + P_t^{(4)}$, is the fundamental driver of the strategy. As shown in Figure 4, this residual series exhibits a powerful mean-reverting, quasi-sinusoidal pattern, ideal for high-frequency trading. The oscillations around the zero mean provide numerous trading opportunities.

2 Trading Strategy & Implementation

The strategy is to trade on the divergence and subsequent convergence of the spread. When S_t widens beyond a statistical threshold τ, we initiate a portfolio of trades that will profit when it reverts to its mean.

1. **Short the Spread ($S_t > \tau$):** If Asset 7 is overvalued, we **SELL** Asset 7, **BUY** Asset 2, and **BUY** Asset 4.
2. **Long the Spread ($S_t < -\tau$):** If Asset 7 is undervalued, we **BUY** Asset 7, **SELL** Asset 2, and **SELL** Asset 4.

This structure ensures the strategy is **market-neutral**, deriving profit from relative mispricing, not market direction.

4 The Stochastic Residual Spread Method for Assets 3 & 6

Based on: *A New Approach to Modeling and Estimation for Pairs Trading*

1 Introduction

We have applied a **Stochastic Residual Spread** model, a sophisticated pairs trading strategy to Asset 3 and Asset 6, two securities exhibiting a high historical correlation of approximately 85%. The model elevates pairs trading from simple heuristics to a robust econometric framework by: 1) defining the spread as a **residual in returns** after accounting for systematic market risk, and 2) using a **State-Space Model** and the **Kalman Filter** to estimate the unobservable (latent) true mispricing, systematically separating the trading signal from market noise.

2 The Conceptual Framework & Mathematical Model

The strategy adapts the Arbitrage Pricing Theory (APT) to define a relative equilibrium between two assets. The core idea is that the return difference between Asset A and Asset B should be explained by their different sensitivities (Γ) to market risk factors (r_t^m). Any unexplained portion is the residual mispricing (e_t) we aim to trade.

$$R_t^A - R_t^B = \Gamma r_t^m + e_t \tag{2}$$

The strategy models this unobservable residual spread, now denoted x_t, as a mean-reverting stochastic process using a State-Space framework.

The State-Space Equations

The model consists of two core equations:

1. **Transition Equation (The Hidden State):** The true mispricing x_t is assumed to follow a **Vasicek process**, which mathematically describes its tendency to revert to a long-run mean θ at a speed κ.

$$dx_t = \kappa(\theta - x_t)dt + \sigma dB_t \tag{3}$$

2. **Measurement Equation (The Observation):** We cannot see x_t directly. We observe the return differential $y_k = R_k^A - R_k^B$, which is linked to the hidden state via the following equation, where Γ is estimated as a parameter:

$$y_k = x_k + \Gamma r_k^m + H\omega_k \tag{4}$$

Here, H is the standard deviation of the measurement noise ω_k. This formulation allows the powerful Kalman Filter to be applied.

3 Estimation Filtering via the Kalman Filter

The **Kalman Filter** is an optimal recursive algorithm for estimating the hidden state x_k from the noisy measurements y_k. It serves two purposes: 1) providing prediction errors for **Maximum Likelihood Estimation (MLE)** of the model parameters $\Psi = \{\theta, \kappa, \sigma, \Gamma, H\}$ on a rolling basis, and 2) extracting the real-time mispricing signal for trading. The filter operates in a continuous two-step **Predict-Update** cycle to refine its estimate with each new data point.

$$\text{Predict:} \qquad x_{k|k-1} = \theta(1 - A) + Ax_{k-1|k-1} \tag{5}$$

$$\text{Update:} \qquad x_{k|k} = x_{k|k-1} + K_k(y_k - (x_{k|k-1} + \Gamma r_k^m)) \tag{6}$$

where K_k is the Optimal Kalman Gain that intelligently blends the prediction with the new information.

4 Trading Rule Formulation Signal Generation

The raw, filtered estimate of the latent spread, x_k, represents the best mathematical guess of the true mispricing. A robust trading strategy requires a layer of logic to translate this estimate into high-conviction signals. Our implementation achieves this through an adaptive, multi-stage process. First, a moving average is applied to the filtered spread (x_{ma}) to smooth high-frequency noise. Second, the recent volatility of this smoothed spread (σ_{ma}) is calculated to create adaptive entry and exit thresholds.

Signal Generation Logic

The strategy generates a position signal based on where the smoothed mispricing (x_{ma}) stands relative to its long-term mean (θ) and its own recent volatility (σ_{ma}). The entry threshold is $\text{Thresh}_{\text{Entry}} = \text{Multiplier} \times \sigma_{ma}$, and the exit threshold is $\text{Thresh}_{\text{Exit}} = \text{Fraction} \times \text{Thresh}_{\text{Entry}}$.

+1 **LONG SPREAD (Buy A, Sell B)**
Signal **+1** is generated for Asset A when the spread is significantly below its mean.
Condition: $x_{ma} < \theta - \text{Thresh}_{\text{Entry}}$

-1 **SHORT SPREAD (Sell A, Buy B)**
Signal **-1** is generated for Asset A when the spread is significantly above its mean.
Condition: $x_{ma} > \theta + \text{Thresh}_{\text{Entry}}$

0 **FLAT / EXIT POSITION**
Signal **0** is generated to close any open position when the spread reverts towards its mean.
Condition: $|x_{ma} - \theta| < \text{Thresh}_{\text{Exit}}$

5 Momentum - Equilibrium Strategy for Asset 5

1 Microstructure Signal Extraction

The process begins by transforming raw market data into a clean, predictive signal. Trade logs are first consolidated by timestamp into single records with a volume-weighted average price. The core predictive feature is then extracted: the **Order Book Imbalance (OBI)**, which quantifies instantaneous supply and demand. For L book levels, it is:

$$\text{OBI}_L = \frac{\sum_{i=1}^{L} \text{BidSize}_i - \sum_{i=1}^{L} \text{AskSize}_i}{\sum_{i=1}^{L} \text{BidSize}_i + \sum_{i=1}^{L} \text{AskSize}_i}$$

A positive OBI signals buying pressure; negative signals selling pressure.

2 Predictive Candlestick Synthesis

This is the strategic core of the data processing pipeline. Instead of using standard time bars, the strategy constructs its own forward-looking candlesticks via `make_candles_simple`. Each component is engineered to embed the predictive OBI signal, creating a richer data structure for the downstream indicators.

- **Open:** Set to the VWAP from the trade log of the prior period. This establishes a baseline price before the new microstructure information is incorporated.
- **High & Low:** These are not historical highs and lows, but rather dynamic liquidity boundaries. They are calculated as the volume-weighted average price of the top 3 ask and bid levels, respectively. This represents the immediate price range defined by available liquidity, serving as a proxy for near-term support and resistance.

$$\text{High} = \frac{\sum_{i=1}^{3} (\text{AskPrice}_i \times \text{AskSize}_i)}{\sum_{i=1}^{3} \text{AskSize}_i}$$

- **Close:** The Close is fundamentally predictive. It is set to the best bid price if the OBI signal is bullish (direction $= 1$) or the best ask price if the signal is bearish (direction $= -1$) and if neither, it is the mid-price.
- **Volume:** The volume is also signal-aware. It is calculated as the sum of liquidity on the side of the market indicated by the OBI signal (e.g., sum of bid sizes for a bullish signal). This quantifies the conviction of the imbalance in terms of market depth.

This synthesis process transforms raw data into a sophisticated analytical input that carries both historical context and predictive intelligence.

3 Signal Generation Engine

The engine translates the proprietary candle data into a sized, actionable trade by integrating two analytical frameworks.

Core Analytical Frameworks

- **Equilibrium Framework:** A multi-faceted system for identifying trend, momentum, and dynamic support/resistance. Its components are:
  - *Momentum Conductor:* $(\max(\text{High}_9) + \min(\text{Low}_9))/2$
  - *Equilibrium Baseline:* $(\max(\text{High}_{26}) + \min(\text{Low}_{26}))/2$
  - *Alpha & Beta Projection Lines:* Two forward-projected lines that form the dynamic **Equilibrium Zone**.

Final Signal Logic and Conviction Scoring

The final trade decision is a multi-step process combining the outputs of the core frameworks.

1. **Directional Signal (`Signal`):** The primary trade direction is determined by the interaction between the Momentum Conductor and the Equilibrium Baseline.
   - A **Buy Signal (+1)** is generated when the Momentum Conductor crosses above the Equilibrium Baseline.
   - A **Sell Signal (-1)** is generated when the Momentum Conductor crosses below the Equilibrium Baseline.
   - A **Neutral Signal (0)** is registered when there is no crossover.
2. **Conviction Score (`Strength`):** A continuous score from 0 to 1 is calculated to measure the confidence behind the signal. It's based on a metric known as Equilibrium Strength:
   - **Equilibrium Strength:** Measures the trend's force by the price's normalized distance from the Equilibrium Zone.

   The final score is: $\text{Strength}_{\text{Final}} = (\text{Strength}_{\text{Eq}})$
3. **Position Sizing:** The final trade volume is a direct function of the conviction score, scaled by a maximum position limit, ensuring that higher-conviction signals receive more capital.

$$\text{Volume} = \text{floor}(\text{Strength}_{\text{Final}} \times \text{max_pos})$$

6 Methodology: The Backtesting Engine

1 The Simulation Kernel

The simulation operates through a discrete temporal progression, where each time-step constitutes an atomic event for state evaluation and potential action.

State Vector Initialization

The system is initialized with a multi-asset historical data manifold. For each asset i, the required inputs are:

- **Limit Order Book ($\mathcal{L}_t^i$):** A chronological sequence of Level-3 snapshots.
- **Alpha Signal Vector (α_t^i):** The strategy's directional signal, where $\alpha_t^i \in \{+1, 0, -1\}$.
- **Position Sizing Scalar ($\mathcal{V}_t^i$):** The target transaction volume.

The kernel dynamically tracks the portfolio's evolving state variables: position exposure (p_t^i) and the mark-to-market performance (PnL_t^i).

2 Microstructure-Aware Execution Protocol

The distinction of this engine lies in its sophisticated execution logic, which simulates the critical process of liquidity consumption and models market impact.

Order Routing Liquidity Consumption

Upon receiving a non-neutral alpha signal, the execution module simulates a market order by "walking the book," sensitive to available depth and position constraints ($[-50, +50]$ units).

- **Long Entry ($\alpha_t^i = +1$):** Consumes liquidity from the **ask side** of the order book.
- **Short Entry ($\alpha_t^i = -1$):** Consumes liquidity from the **bid side**.

The executable volume v_{exec} at each price level is a constrained optimization:

$$v_{\text{exec}} = \min(\mathcal{V}_{\text{remaining}}, \mathcal{Q}_{\text{level}}, \text{Capacity}_{\text{limit}})$$

Performance Attribution Cost Modeling

Profit and Loss is updated instantaneously upon each partial fill, incorporating a transaction cost parameter c_{txn} to model fees and execution slippage. The state update function for the PnL is defined as:

- **For a buy at P_{ask}:** $PnL_t = PnL_{t-1} - P_{\text{ask}} \times v_{\text{exec}} \times (1 + c_{txn})$
- **For a sell at P_{bid}:** $PnL_t = PnL_{t-1} + P_{\text{bid}} \times v_{\text{exec}} \times (1 - c_{txn})$

This rigorous, fill-by-fill accounting provides a precise trajectory of the strategy's profitability.

3 Order Fill & Volume Reconciliation (`Volume Fix`)

To bridge the gap between theoretical signals and executable trades, the engine incorporates a volume reconciliation function. The function operates in two primary modes:

- **Unfilled Volume Carry-Forward:** After an execution attempt, the engine compares the executed volume against the target volume dictated by the alpha signal. Any unfilled portion of the original order is not discarded; instead, it is carried forward as a new pending order for the subsequent time-step.
- **Position Squaring and Reversal:** The function continuously reconciles the current actual position with the new target position from the latest signal. If a signal change necessitates reducing or reversing an existing position (e.g., a long signal flips to neutral or short), the function automatically generates the requisite closing or reversing trades to square-off the now-unwanted exposure.

4 Risk Management Protocol

To ensure capital preservation and enforce discipline, the engine integrates a dynamic, percentage-based stop-loss mechanism. The system maintains a record of the entry price (P^i_{entry}) for each asset i, and a pre-configured stop-loss percentage (δ_{SL}) defines the maximum tolerable adverse price movement.

- **Long Position Stop-Loss:** For a long position, a stop-loss is triggered if the current market price (P^i_t) falls below the entry price by the specified percentage. This initiates an immediate market sell order to liquidate the position.

  $$\text{Trigger Condition: } P^i_t \leq P^i_{\text{entry}} \times (1 - \delta_{SL})$$

- **Short Position Stop-Loss:** For a short position, the trigger occurs if the current price rises above the entry price by the same percentage, necessitating a market buy order to cover the position and cap the loss.

  $$\text{Trigger Condition: } P^i_t \geq P^i_{\text{entry}} \times (1 + \delta_{SL})$$

This rule-based approach systematically prevents catastrophic drawdowns by ensuring that no single trade can incur a loss beyond a strictly defined risk parameter.

5 Terminal State

Mandated Position Liquidation

At the simulation's conclusion ($t = T$), a mandatory liquidation protocol is initiated to close all open exposures. Residual long positions are sold into the final bid book, and short positions are covered from the final ask book at the best available price. This ensures the total PnL is a true representation of round-trip performance.

7 Results of Backtesting

```
Total PNL: 1419445.800190007
PNL_Asset1: 319067.77604000224
PNL_Asset2: 44600.23910000027
PNL_Asset3: 113655.1037400001
PNL_Asset4: 54206.35790999996
PNL_Asset5: 702892.7472800042
PNL_Asset6: 95531.10394000025
PNL_Asset7: 89492.47218000004
```

Figure 5: Cumulative PnL of the Multi-Strategy Portfolio